

EDUKIT

School Commerce, Marketplace & Fulfilment Management System

SOFTWARE REQUIREMENTS SPECIFICATION

Version 2.0

Project Name:	EduKit
Document Version:	2.0
Document Status:	Revised Requirements Specification
Primary Market:	Uganda
Primary Currency:	Uganda Shillings (UGX)
Technology Baseline:	Laravel 12, PHP 8.3+, Jetstream, Livewire, Spatie Permission

Parents order. Suppliers fulfil. Drivers deliver. Schools or recipients confirm.

Prepared for requirements validation and development planning.

Document Control

Version	Status	Summary
1.0	Draft	Initial EduKit Store SRS covering e-commerce, payments, suppliers, delivery, wallet and reporting.
2.0	Revised	Expanded into a hybrid online shop, supplier marketplace, sourcing engine, delivery network, school receiving network and settlement platform.

Purpose of this Document

This document defines the functional, operational, financial, integration, security and technical requirements for EduKit. It is intended to guide system design, implementation, testing, stakeholder validation and future enhancement planning.

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1. Introduction

EduKit is a multi-vendor school-commerce, shopping-list fulfilment, sourcing, delivery and financial-management platform. It enables parents and guardians to purchase school-related products online, upload shopping lists, associate purchases with students and schools where applicable, select a delivery destination, pay digitally, and track fulfilment to verified receipt.

EduKit may fulfil an order from its own inventory or from approved third-party suppliers. The complete lifecycle is: customer request, product selection or shopping-list matching, cart, checkout, payment, supplier selection, supplier acceptance, preparation, driver assignment, pickup, delivery, recipient confirmation, financial settlement and order completion.

2. Business Vision

EduKit shall operate as a hybrid online shop, supplier marketplace, procurement engine, delivery network, school receiving network and financial platform. Customers should experience EduKit as one trusted platform even when fulfilment is performed by different suppliers and delivery partners.

3. Business Objectives

- Digitise purchasing of school requirements in Uganda.
- Allow parents and guardians to purchase products for one or more children.
- Associate school-bound orders with a specific school and student.
- Support school, home, alternative-address and self-pickup fulfilment.
- Allow customers to upload school shopping lists instead of searching item by item.
- Maintain a controlled master product catalogue.
- Allow approved suppliers to register the products they supply, supplier cost, stock and availability.
- Recommend the best supplier using product coverage, cost, distance, preparation time and performance.
- Support semi-automated supplier assignment at launch and optional full automation later.
- Allow suppliers to accept, decline, prepare and hand over orders.
- Assign delivery jobs to eligible delivery partners.
- Calculate customer delivery fees and driver payouts separately.
- Allow schools and other recipients to independently confirm receipt.
- Manage supplier and driver settlements, refunds, wallets and financial ledgers.
- Provide operational, financial and performance dashboards and reports.

4. System Actors and Roles

4.1 Super Administrator

- Full system access and configuration.
- Manage roles, permissions, pricing rules, delivery rules, supplier rules, settlement rules and integrations.
- Access complete audit information.

4.2 EduKit Administrator

- Manage products, categories, schools, students, suppliers, orders, shopping lists, delivery operations and notifications.
- Review supplier recommendations and override them when authorised.
- Manage operational exceptions.

4.3 Finance Administrator

- Manage customer payments, supplier payables, driver payables, refunds, wallets and reconciliations.
- Review gross contribution, service fees, logistics margin and settlement history.

4.4 Parent / Customer

- Register and manage account details.
- Create student profiles and saved delivery addresses.
- Browse products or upload a shopping list.
- Checkout, pay, track orders, view receipts and receive notifications.

4.5 Supplier

- Maintain business profile, products, supplier prices, stock and availability.
- Receive, accept or decline order requests.
- Prepare orders, mark ready for pickup and view payments and performance.

4.6 Delivery Person

- Maintain delivery profile and operating area.
- Receive eligible delivery jobs, accept or decline, navigate to pickup and destination, update status and view earnings.

4.7 School Administrator / School Receiver

- View deliveries destined for the authorised school.
- Search or scan orders and verify student/order details.
- Confirm full or partial receipt and report delivery discrepancies.

4.8 Role-Based Access Control

Laravel shall use Spatie Laravel Permission. Initial roles include super-admin, admin, finance-admin, supplier, delivery-person, school-admin, school-receiver and parent. Permissions shall be granular and enforced at both route/controller and record-ownership level.

5. High-Level System Components

- Public e-commerce website for browsing, supplier registration, delivery-partner registration, shopping-list upload and order tracking.

- Customer portal for profiles, cart, checkout, payment, orders, wallet, receipts and notifications.
- EduKit administration dashboard for complete operational and financial management.
- Supplier portal for offers, prices, stock, orders, fulfilment and payouts.
- School portal for incoming deliveries, lookup, receipt confirmation and issue reporting.
- Delivery application or mobile-friendly portal for jobs, navigation, pickup, delivery and earnings.
- Central Laravel backend/API for authentication, business logic, supplier selection, payments, delivery assignment, financial ledger, reporting and notifications.

6. Customer Registration and Authentication

- Register using phone number and/or email address.
- Phone OTP verification and email verification where applicable.
- Secure password/PIN handling and account recovery.
- Profile update and saved addresses.
- Jetstream/Fortify for web authentication and Sanctum for API/mobile authentication.

7. Student Profile Management

A student shall not require a phone number or independent EduKit account. The parent or guardian remains the account holder.

- Student full name.
- School.
- Class and stream.
- Admission/student number where applicable.
- Boarding/day status where applicable.
- Delivery notes.

8. School Management

- School name and code.
- School type.
- District and physical address.
- Latitude and longitude.
- Contact information.
- Active/inactive status.
- Delivery instructions.
- Supported delivery days where applicable.
- Multiple authorised receiving users.

9. Master Product Catalogue

The master catalogue shall be controlled by EduKit administrators to prevent duplicate and inconsistent product naming across suppliers.

- Product name, SKU, category and description.
- Images, brand, variant, size and unit.
- Customer selling price.
- Availability status.
- School applicability where relevant.
- Categories such as books, stationery, uniforms, shoes, bags, bedding, toiletries, laboratory items, sports equipment and other school requirements.

10. Supplier Product Model

Suppliers shall not freely create duplicate catalogue products. A supplier selects an existing master product and indicates that it is supplied by the business. If the required product does not exist, the supplier may submit a new-product request for administrator approval.

- Supplier and product reference.
- Current supplier cost.
- Available stock quantity.
- Availability status.
- Minimum order quantity.
- Preparation/lead time.
- Supplier location.
- Effective pricing date.

11. Supplier Price History

Updating a supplier price shall never alter the supplier cost already captured on historical orders.

- Supplier.
- Product.
- Cost price.
- Valid-from date.
- Valid-until date.
- Created-by user.
- Timestamp.

12. Customer Selling Price

The customer normally sees an EduKit selling price rather than the supplier cost. Supplier cost and customer selling price are independent values so that EduKit can manage margins consistently while sourcing from the most suitable supplier.

A future marketplace mode may expose multiple supplier offers to customers, but this is outside the initial fulfilment model.

13. EduKit Own Inventory

EduKit may act as a fulfilment source itself. Every fulfilment source shall be classified as `EDUKIT_STOCK` or `THIRD_PARTY_SUPPLIER`. Supplier selection may consider EduKit-owned inventory together with approved external supplier offers.

14. Shopping Cart

Cart subtotal equals the sum of customer selling price multiplied by quantity for every item.

- Add and remove products.
- Increase or decrease quantities.
- View price and subtotal.
- Save cart where supported.
- Proceed to checkout.

15. Shopping List Upload

Customers may upload or photograph a school shopping list instead of browsing item by item. The upload shall create a Shopping List Request that can be matched to the master catalogue.

- Take a photo.
- Upload an image.
- Upload a PDF or supported document.
- Identify or extract requested items.
- Match items to catalogue products.
- Confirm quantities.
- Allow administrator/customer corrections.
- Require customer review before the matched items can be paid for.

16. Checkout

- Customer.
- Products and quantities.
- Student and school where applicable.
- Recipient.
- Delivery type and destination.
- Additional notes.
- Delivery fee.
- Service/convenience fee.
- Payment method.
- Total payable.

17. Delivery Destination Types

17.1 School Delivery

- School.
- Student.
- Class/stream.
- Student or admission number where available.

17.2 Parent / Home Delivery

- Recipient name.
- Phone number.
- District and area.
- Physical address and map pin where possible.

17.3 Other Address

- Guardian, relative, office, hostel or another authorised recipient.
- Recipient name, phone, address and location details.

17.4 Self Pickup

The customer chooses an available collection location. Normal delivery fees do not apply unless a pickup/handling charge is explicitly configured.

18. Service / Convenience Fee

EduKit shall support configurable service fees displayed separately at checkout.

- Fixed fee.
- Percentage fee.
- Minimum/maximum fee.
- School-specific fee.
- Area-specific fee.
- Order-value based fee.

19. Delivery Pricing

Customer delivery fees shall be configurable. A typical calculation may combine base charge, distance charge, additional pickup charges, package/weight charge and special handling.

- District or city-zone pricing.
- Inter-district pricing.
- Scheduled school-delivery pricing.

- Express-delivery pricing.
- Free-delivery promotions.

20. Customer Delivery Fee vs Driver Payout

The customer delivery fee and driver payout shall be stored as separate financial values. The difference may represent a logistics margin or may be zero depending on platform configuration.

- customer_delivery_fee
- driver_payout
- other direct delivery costs
- logistics margin

21. Payment Management

EduKit shall not collect or store a customer Mobile Money PIN.

- MTN Mobile Money through the configured payment gateway.
- Airtel Money through the configured payment gateway.
- Visa/card.
- EduKit Wallet where enabled.
- Server-side payment verification.
- Webhook processing with duplicate protection.
- Statuses: Pending, Successful, Failed, Cancelled, Reversed and Refunded.

22. Order Creation

- Unique order number such as EDK-2026-000125.
- Customer, student and school where applicable.
- Recipient and delivery type.
- Order items and quantities.
- Historical customer selling price.
- Historical supplier cost.
- Subtotal, service fee, delivery fee and total.
- Payment reference/status.
- Supplier allocation and delivery assignment.
- Complete status history.

23. Order Status Workflow

Exception states include Payment Failed, Supplier Declined, Supplier Timeout, Partially Fulfilled, Partially Received, Missing Items, Damaged Items, Delivery Disputed, Cancelled, Returned and Refunded.

- Pending Payment.

- Payment Successful.
- Order Confirmed.
- Supplier Selection.
- Awaiting Supplier Acceptance.
- Supplier Accepted.
- Preparing.
- Ready for Pickup.
- Driver Assignment.
- Driver Assigned.
- Picked Up.
- In Transit.
- Arrived at Destination.
- Awaiting Recipient Confirmation.
- Delivered.
- Delivery Confirmed.
- Completed.

24. Supplier Recommendation Engine

Supplier recommendation shall initially be rule-based and deterministic. AI is not required for the first implementation. The engine shall first filter ineligible suppliers and then rank eligible suppliers.

25. Supplier Eligibility Filter

- Supplier is approved.
- Supplier account is active.
- Supplier is accepting orders.
- Supplier provides relevant products.
- Supplier has sufficient stock.
- Supplier operates in a supported sourcing area.
- Supplier is not suspended.
- Supplier satisfies administrator-defined eligibility rules.

26. Whole-Order Product Coverage

For each eligible supplier, EduKit shall calculate the percentage or count of order lines and required quantities that can be fulfilled. The system shall prefer complete-order fulfilment where economically reasonable because unnecessary multi-supplier splitting increases pickup and delivery cost.

27. Supplier Goods Cost

Supplier Goods Cost = Sum of valid supplier price multiplied by the required quantity for all products allocated to that supplier. Only prices valid at the time of allocation shall be used.

28. Landed Fulfilment Cost

Landed Cost = Supplier Goods Cost + Estimated Pickup/Logistics Cost. This prevents the platform from selecting a supplier solely because of low unit prices when the transport required to use that supplier would make the total order more expensive.

29. Supplier Scoring

Initial scoring weights shall be configurable. A recommended starting point is product coverage 40 percent, landed cost 25 percent, distance/location 20 percent, preparation time 10 percent and supplier performance 5 percent. New suppliers without sufficient history shall receive a neutral performance score.

30. Supplier Performance

Performance metrics may progressively influence supplier ranking after sufficient real transaction history exists.

- Requests received.
- Acceptance rate.
- Decline rate.
- Timeout rate.
- Fulfilment rate.
- Preparation time.
- Stock accuracy.
- Cancellation rate.
- Damaged/incorrect item complaints.
- Successful delivery rate.

31. Semi-Automated Supplier Assignment

31.1 Initial Rollout Flow

1. Confirm customer payment.
2. Filter eligible suppliers.
3. Calculate whole-order coverage.
4. Calculate supplier goods cost.
5. Estimate logistics and landed cost.
6. Calculate score.
7. Rank suppliers.
8. Show recommendation to EduKit administrator.
9. Administrator approves or overrides.
10. Selected supplier receives an order request.

32. Fully Automated Supplier Assignment

After the recommendation logic has been validated with sufficient operational data, authorised administrators may enable automatic assignment. The highest-ranked eligible supplier receives the first request automatically, with admin intervention reserved for exceptions.

33. Supplier Order Request

- Order reference.
- Required products and quantities.
- Expected supplier payout.
- Required preparation deadline.
- Pickup requirements.
- Accept or Decline action.
- Decline reasons such as out of stock, price changed, insufficient quantity, cannot meet deadline, temporary closure or other.

34. Supplier Response Time

Supplier order requests shall have a configurable response window. If a supplier declines or does not respond before expiry, the request may automatically fall back to the next ranked eligible supplier. Every attempt shall be logged.

35. Split Order Fulfilment

If no single supplier can fulfil the order, EduKit may split it among multiple suppliers. Before splitting, the platform shall consider product cost, number of pickup points, additional logistics cost, preparation time and customer delivery deadline. Split fulfilment shall not be selected merely because individual items are cheaper.

36. Supplier Portal

- Dashboard.
- My Products.
- My Prices.
- Stock and Availability.
- Order Requests.
- Accepted/Preparing Orders.
- Ready for Pickup.
- Completed Orders.
- Payments and Outstanding Balance.
- Performance.
- Business Profile and Notifications.

37. Delivery Partner Onboarding

- Name and phone.
- Identity details and photo where required.
- Operating district/area.
- Vehicle type.
- Vehicle licence/registration details where relevant.
- Payment/mobile-money details.
- Availability status.
- Administrator approval before receiving jobs.

38. Delivery Job Assignment

The system shall first filter eligible delivery persons and then rank them. Eligibility may consider approved status, availability, operating area, pickup location, destination, vehicle capability and package requirements. Ranking may consider distance to pickup, delivery distance, reliability, acceptance rate and current workload.

39. Delivery Job Offer

If a driver declines or times out, the job may be offered to another eligible driver.

- Pickup supplier and address.
- Destination.
- Approximate route.
- Package count.
- Estimated driver payout.
- Special instructions.
- Accept or Decline.

40. Delivery Workflow

Assigned -> Travelling to Pickup -> Arrived at Pickup -> Package Collected -> In Transit -> Arrived at Destination -> Awaiting Confirmation -> Confirmed -> Completed.

- Driver.
- Supplier.
- Order/package.
- Date/time.
- Optional GPS.
- Optional photo/signature at pickup or delivery.

41. School Delivery Confirmation

School delivery shall use independent recipient confirmation. The driver shall not be the only party able to complete a school-bound delivery.

- Scan delivery QR code or enter order number.
- Search authorised incoming school deliveries.
- Display order number, student, class, admission number where appropriate, products and quantities.
- Confirm All Received.
- Partially Received.
- Missing Item.
- Damaged Item.
- Incorrect Item.
- Reject Delivery.
- Store receiver, school, timestamp, comments and optional signature/photo/GPS.

42. Home / Other Address Confirmation

The exact proof-of-delivery combination shall be configurable.

- OTP confirmation.
- Recipient signature.
- Recipient name.
- Delivery photograph.
- GPS confirmation.
- Manual authorised confirmation.

43. Self-Pickup Confirmation

The collection point or supplier shall mark the order collected after successful verification.

- QR code.
- Pickup code.
- OTP.
- Identification confirmation.

44. Consolidated School Deliveries

EduKit should support grouping multiple parent orders destined for the same school into one master delivery while keeping each parent order financially and operationally separate. Packages should remain individually labelled by student/order. Scheduled school delivery days may be configured to reduce logistics cost.

45. Delivery Person Payment

Driver earnings become payable only after successful delivery confirmation according to configured business rules.

- Delivery reference.
- Gross driver payout.
- Amount paid.
- Outstanding amount.
- Payment date.
- Transaction reference.
- Payment status.

46. Supplier Settlement

Supplier cost is committed when an order is accepted. Payout eligibility may occur after verified pickup or successful final delivery, depending on configured settlement rules. Supported cycles may include per-order, daily, weekly or another approved schedule.

- Supplier cost committed.
- Supplier amount payable.
- Supplier amount paid.
- Outstanding balance.
- Settlement date and reference.

47. EduKit Revenue Model

47.1 Own-Stock Product Margin

Product Margin = Customer Selling Price - EduKit Product Cost.

47.2 Supplier-Fulfilled Product Margin

Marketplace/Fulfilment Margin = Customer Selling Price - Supplier Cost, or a separately configured supplier commission model.

47.3 Service / Convenience Fee

A separately displayed checkout fee may be charged under configured rules.

47.4 Logistics Margin

Logistics Margin = Customer Delivery Fee - Driver Payout - Other Direct Delivery Costs. The margin may be zero.

47.5 Future Revenue Options

- Supplier subscription plans.
- Promoted supplier placement.
- School procurement services.
- Premium/express fulfilment.
- Bulk procurement fees.

48. Financial Ledger

Balances and financial history shall be ledger-based rather than relying only on manually editable balance fields.

- Customer payment.
- Product revenue.
- Supplier cost.
- Supplier payable and supplier settlement.
- Customer delivery charge.
- Driver payable and driver settlement.
- Service fee.
- Payment gateway charge where available.
- Refund.
- Wallet movement.
- EduKit gross contribution.

49. Customer Wallet

Every wallet movement shall create an immutable ledger transaction with references and timestamps.

- Deposit.
- Purchase.
- Refund.
- Credit.
- Debit.
- Adjustment.
- Withdrawal if enabled.

50. Refunds and Returns

Refunds may be returned to the original payment method, EduKit Wallet or another authorised mechanism. All refund activity must be audited.

- Unavailable products.
- Customer cancellation.
- Failed fulfilment.

- Missing items.
- Incorrect items.
- Damaged products.
- Supplier failure.
- Delivery failure.

51. Digital Receipts

- EduKit branding.
- Receipt serial number.
- Order number.
- Customer.
- Student and school where applicable.
- Products and quantities.
- Subtotal.
- Service fee.
- Delivery fee.
- Total paid.
- Payment method and reference.
- Date/time and status.

52. Customer Notifications

Channels may include SMS, email, push, in-app notifications and later WhatsApp Business integration.

- Registration and verification.
- Payment initiated/successful/failed.
- Order confirmed.
- Supplier assigned/preparing/ready.
- Driver assigned.
- Picked up and in transit.
- Arrived at destination.
- Delivered and recipient confirmed.
- Issue reported.
- Refund processed.
- Order completed.

53. Admin Order Entry

EduKit administrators shall be able to create orders received by phone, WhatsApp, walk-in or other approved offline channels. Manual/admin orders shall be clearly identified and shall proceed through the normal sourcing, payment and fulfilment workflow subject to authorisation rules.

54. Admin Dashboard

- Total and today's orders.
- Revenue.
- Pending payments.
- Orders awaiting supplier.
- Supplier requests awaiting response.
- Preparing and ready-for-pickup orders.
- Deliveries assigned/in transit/awaiting confirmation.
- Completed deliveries.
- Failed payments and refunds.
- Supplier payables.
- Driver payables.
- EduKit gross contribution.

55. Financial Reports

- Sales by date, school, district, product, category and customer.
- Supplier costs and supplier settlements.
- Driver costs and delivery revenue.
- Platform fees.
- Payment-method reports.
- Profitability and gross contribution.
- Refunds.
- Wallet balances.
- Outstanding payables.

56. Supplier Reports

- Supplier fulfilment value.
- Product costs.
- Accepted, declined and timed-out requests.
- Fulfilment rate.
- Preparation time.
- Stock accuracy.
- Outstanding payment and payment history.
- Profitability contribution.

57. Delivery Reports

- Delivery volume.
- Driver performance.
- Completed and failed deliveries.
- Average delivery time.
- Distance.
- Delivery revenue.
- Driver payouts.
- Logistics margin.
- Outstanding payments.

58. School Reporting

School users shall only see data for schools to which they are explicitly authorised.

- Orders by school.
- Student packages.
- Delivery success rate.
- Outstanding confirmations.
- Delivery issues.
- Aggregated school order volume.

59. Search and Filtering

- Search by order number, customer, phone, student, admission number, school, supplier, driver, product and payment reference.
- Filter by date, district, school, supplier, driver, order status, payment status, delivery status, order source and delivery type.

60. Audit Log

Audit records should store user, action, affected entity, previous value, new value, timestamp and IP/device information where appropriate.

- Product and catalogue changes.
- Customer selling-price changes.
- Supplier-price changes.
- Supplier approval.
- Supplier-assignment override.
- Driver-assignment override.
- Stock changes.
- Payment status changes.

- Refunds and wallet adjustments.
- Delivery confirmation.
- Settlement actions.
- Role and permission changes.

61. Location and Maps

- Schools.
- Supplier locations.
- EduKit warehouses.
- Pickup points.
- Customer addresses.
- Delivery-person locations/operating areas.
- Coordinates and distance calculations.
- Nearby supplier matching.
- Nearby driver matching.
- Navigation and route estimates.

62. Core Business Rules

- BR-001: A normal order shall not enter supplier fulfilment until payment is successfully verified.
- BR-002: Historical customer and supplier prices attached to an order shall never change because catalogue prices later change.
- BR-003: Only approved suppliers may receive order requests.
- BR-004: Supplier selection shall first exclude ineligible suppliers.
- BR-005: Whole-order fulfilment is preferred over unnecessary multi-supplier splitting.
- BR-006: Cheapest product price alone shall not determine supplier selection.
- BR-007: Estimated logistics and landed cost shall be considered.
- BR-008: An authorised administrator may override supplier recommendations.
- BR-009: New suppliers receive neutral performance scoring until sufficient history exists.
- BR-010: Supplier requests may expire and proceed to another supplier.
- BR-011: A driver may only receive jobs for which the driver is eligible and available.
- BR-012: Customer delivery fee and driver payout are separate financial values.
- BR-013: School deliveries require authorised receiving confirmation before normal completion.
- BR-014: Home and other-address deliveries require configured recipient proof.
- BR-015: Driver payout becomes payable only after successful delivery confirmation.
- BR-016: Financial adjustments require ledger and audit records.
- BR-017: A school user shall only access authorised school orders.
- BR-018: A supplier shall only access its own products, orders and financial records.
- BR-019: Shopping-list interpretation requires customer validation before payment/order confirmation.
- BR-020: The platform shall retain a complete order-status history.

63. Core Database Entities

63.1 Identity

- users
- roles
- permissions

63.2 Customers

- customers
- customer_addresses
- students

63.3 Schools

- schools
- school_users
- school_receivers
- districts

63.4 Catalogue

- product_categories
- products
- product_variants
- product_images
- product_prices

63.5 Suppliers

- suppliers
- supplier_users
- supplier_locations
- supplier_products
- supplier_prices
- supplier_performance_metrics

63.6 Shopping Lists

- shopping_lists
- shopping_list_items
- shopping_list_matches

63.7 Orders

- orders
- order_items
- order_status_history
- order_allocations

63.8 Supplier Fulfilment

- supplier_order_requests
- supplier_orders
- supplier_order_items
- supplier_request_history

63.9 Payments

- payments
- payment_transactions
- payment_webhooks
- receipts
- refunds

63.10 Delivery

- delivery_persons
- driver_availability
- delivery_jobs
- delivery_assignments
- delivery_stops
- delivery_status_history
- delivery_confirmations
- delivery_payments

63.11 Finance

- financial_ledger_entries
- supplier_payments
- supplier_settlements
- delivery_payments
- platform_fees
- commissions
- settlements

63.12 Wallet

- wallets
- wallet_transactions

63.13 Communication

- notifications
- notification_templates
- notification_logs

63.14 Administration

- audit_logs
- system_settings
- fee_rules
- delivery_rate_rules

64. API Requirements

64.1 Authentication

- Register, login, logout, OTP verification and password reset.

64.2 Catalogue

- Categories, products, product details and search.

64.3 Customer

- Student profiles, addresses and shopping lists.

64.4 Cart

- Add, remove, update and calculate totals.

64.5 Orders

- Create, view, cancel and track orders.

64.6 Payments

- Initiate, verify, webhook handling and refund.

64.7 Suppliers

- Profile, products, prices, stock, order requests, accept/decline and preparation status.

64.8 Deliveries

- Available/assigned jobs, accept/decline, pickup, status, proof of delivery and earnings.

64.9 Schools

- Incoming deliveries, order lookup, receipt confirmation and discrepancy reporting.

64.10 Administration

- Users, products, schools, suppliers, orders, supplier recommendations, deliveries, finance, reports and settings.

65. Recommended Technology Architecture

65.1 Backend

- Laravel 12.
- PHP 8.3 or later compatible release.
- REST API.

65.2 Web Authentication

- Laravel Jetstream.
- Laravel Fortify.
- Laravel Sanctum.

65.3 Roles and Permissions

- Spatie Laravel Permission.

65.4 Web Frontend

- Blade.
- Livewire.
- Tailwind CSS.

65.5 Database

- PostgreSQL or MySQL.

65.6 Mobile

- Flutter customer application in a later phase.
- Flutter delivery application in a later phase.
- Responsive web/PWA may be used during MVP where appropriate.

65.7 Infrastructure

- Linux cloud hosting.
- Managed database where possible.
- Redis/cache.
- Queue workers.
- Object storage.
- Monitoring and backups.

65.8 Integrations

- Flutterwave or configured payment provider.
- SMS provider.
- Email provider.
- Maps/geolocation provider.
- Push notifications.
- WhatsApp Business API in a later phase.

66. Security Requirements

- HTTPS/TLS.
- Secure password hashing.
- Role-based permissions.
- API authentication.
- Input validation.
- CSRF protection.
- SQL injection protection.
- XSS protection.
- Rate limiting.
- Audit logging.
- Secure payment handling.
- Server-side payment verification.
- Least-privilege access.
- Record ownership and tenant-like scoping for suppliers and schools.

67. Privacy Requirements

- Collect only information required for fulfilment and support.
- Restrict student information.
- Restrict school order information.
- Limit driver access to delivery-essential information.

- Protect parent/guardian data.
- Define retention, archiving and deletion rules.
- Log sensitive administrative access where appropriate.

68. Performance Requirements

- Normal API responses should target approximately 1-3 seconds under normal expected load.
- Supplier selection shall remain efficient as supplier volume increases.
- Large reporting processes should use queued/background jobs where appropriate.
- Images and uploaded shopping lists should be stored outside the primary relational database.

69. Availability and Reliability

- Automated application and database backups.
- Error monitoring and structured logs.
- SSL management.
- Queue workers and retry mechanisms.
- Failed-job monitoring.
- Idempotent payment webhooks.
- Separate staging and production environments.
- Documented disaster-recovery procedures.

70. Acceptance Criteria

70.1 AC-001 Customer Purchase

- Successful payment is verified.
- Unique order number is generated.
- Receipt is generated.
- Supplier sourcing begins.

70.2 AC-002 Supplier Recommendation

- Ineligible suppliers are excluded.
- Product coverage is calculated.
- Supplier cost and landed cost are calculated.
- Supplier score is calculated.
- Eligible suppliers are ranked.

70.3 AC-003 Supplier Failure

- Decline/timeout is recorded.
- Next eligible supplier may receive the request.
- Customer order remains active unless an exception requires intervention.

70.4 AC-004 School Delivery

- Driver cannot independently finalise school receipt.
- Authorised school receiver confirms or disputes the delivery.

70.5 AC-005 Home Delivery

- Configured recipient proof is captured before completion.

70.6 AC-006 Supplier Pricing

- Current price changes do not alter previous order profitability.

70.7 AC-007 Financial Settlement

- Customer revenue, supplier cost/payable, delivery revenue, driver payable, service fee and EduKit gross contribution are recorded separately.

70.8 AC-008 Shopping List

- Customer reviews matched items before checkout/payment.

70.9 AC-009 RBAC

- Supplier cannot access another supplier data.
- School cannot access another school deliveries.
- Driver cannot access unrelated customer orders.

71. MVP - Phase 1

- Laravel 12 foundation.
- Jetstream authentication.
- Spatie roles/permissions.
- Public website and customer portal.
- Student profiles.
- Master product catalogue.
- Cart and checkout.
- Multiple delivery destinations.
- Flutterwave payment integration.
- Orders and receipts.
- Supplier onboarding and approval.
- Supplier Portal.
- Supplier products, prices, stock and availability.
- Semi-automated Supplier Recommendation Engine.
- Admin supplier approval/override.

- Supplier order acceptance.
- Delivery-person management and basic assignment.
- School Portal and school receipt confirmation.
- Home-delivery confirmation.
- Notifications.
- Core finance/ledger.
- Admin Dashboard.

72. Phase 2 - Operational Automation

- Automatic supplier fallback.
- Advanced driver matching.
- Dynamic delivery rates.
- Consolidated school deliveries.
- Shopping-list upload workflow.
- Supplier and driver performance scoring.
- Supplier settlement management.
- Driver settlement management.
- Financial reports and advanced dashboards.
- Customer wallet.
- Admin-created WhatsApp/phone orders.

73. Phase 3 - Advanced Platform

- Fully automatic supplier assignment.
- Route optimisation.
- Flutter customer app.
- Flutter delivery app.
- AI-assisted shopping-list interpretation.
- Automated WhatsApp ordering.
- Supplier subscriptions.
- School-specific catalogues.
- School procurement.
- Bulk school orders.
- Promotions and loyalty.
- Automated reconciliation.
- Demand forecasting.
- Advanced analytics.

74. Configuration Requiring Business Approval

These rules should be configurable rather than hard-coded so that EduKit can change commercial policy without redeploying the application.

- EduKit product markup.
- Supplier commission model.
- Service/convenience fee.
- Delivery pricing.
- Driver payout.
- Logistics margin.
- Supplier response timeout.
- Supplier scoring weights.
- Supplier settlement cycle.
- Driver settlement cycle.
- Return and refund rules.
- Wallet withdrawals.
- Cash-on-Delivery availability.
- Supplier verification requirements.
- Driver verification requirements.
- Scheduled school-delivery rules.
- Split-order threshold.

75. Complete System Workflow

75.1 Customer

Register/Login -> Create or select student where applicable -> Browse catalogue or upload shopping list -> Add products -> Cart -> Choose delivery destination -> Review charges -> Pay -> Payment verified -> Order created.

75.2 Sourcing

Order paid -> Find eligible suppliers -> Check stock -> Calculate whole-order coverage -> Calculate supplier cost -> Estimate logistics/landed cost -> Score and rank suppliers -> Recommend supplier -> Admin approval during initial rollout -> Supplier receives request -> Accept/decline/timeout -> Next supplier if required -> Prepare -> Ready for pickup.

75.3 Delivery

Find eligible driver -> Offer job -> Driver accepts -> Travel to supplier -> Collect package -> In transit -> Arrive at destination.

75.4 Confirmation

School: authorised receiver confirms by order/QR; Home/other address: recipient confirms by configured proof; Self-pickup: pickup code/QR confirms handover. Outcomes may be delivered, partially received or disputed.

75.5 Finance

Customer payment -> Supplier cost recorded -> Driver cost recorded -> Service fee and delivery revenue recorded -> Supplier payable -> Driver payable -> EduKit margin calculated -> Supplier/driver settlement -> Reports updated.

76. Final System Definition

EduKit is not merely an e-commerce website. It is a School Commerce, Supplier Marketplace, Procurement, Fulfilment, Delivery and Financial Management Platform connecting parents, students, schools, suppliers, delivery partners and EduKit operations.

Its central value proposition is: Parents order. Suppliers fulfil. Drivers deliver. Schools or recipients confirm. EduKit coordinates and settles the complete transaction.

The platform shall provide a trusted digital chain from a parent or guardian shopping requirement to verified final receipt while enabling EduKit to operate a sustainable business through controlled product margins, supplier fulfilment, service fees and logistics.

Approval / Requirements Validation

Role	Name / Signature	Date
Project Owner		
Business Representative		
Technical Lead		
Finance / Operations		